ISRAEL'S SEQUENCES FOR THE ALTERNATING CONTINUED FRACTION $1-1/(2+1/(3-1/(4+\cdots)))$ ARE COPRIME INTEGERS (OEIS A262957)

MATH0

Abstract. Let $\xi_n = 1-1/(2+1/(3-\cdots(n+1)+(-1)^{n+1}/1))$ and let $a(n)$ be the numerator of ξ_n in lowest terms (OEIS A262957). In 2015 Robert Israel gave a three-term recurrence with rational coefficients for two sequences $b(n), c(n)$ and conjectured that $b(n)$ and $c(n)$ are coprime for all n, so that $a(n) = b(n)$. We prove that for every $n \geq 1$ the numbers $b(n)$ and $c(n)$ are coprime integers and $a(n) = b(n)$; along the way we show that $b(n), c(n) > 0$ and $\xi_n = b(n)/c(n)$. The proof identifies $b(n)$ and $c(n)$ with $P_n \pm P_{n-1}$ and $Q_n \pm Q_{n-1}$, where P_k/Q_k are the convergents, and uses the determinant identity $P_kQ_{k-1} - P_{k-1}Q_k = \pm 1$. To our knowledge the result is new (the entry still lists it as a conjecture), although it is a routine consequence of classical continued-fraction theory once this identification is made. The theorem is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the listed terms; this statement is not part of the formal-conjectures repository.

1. Introduction

For an integer $k \geq 1$ and a rational $t \neq 0$ put $f_k(t) = k + (-1)^k/t$, and for $n \geq 0$ let

$$\xi_n = f_1\big(f_2(\cdots f_{n+1}(1)\cdots)\big) = 1 - \cfrac{1}{2 + \cfrac{1}{3 - \cfrac{1}{\ddots\ (n+1) + \cfrac{(-1)^{n+1}}{1}}}}.$$

Thus $\xi_1 = 2/3$, $\xi_2 = 3/5$, $\xi_3 = 19/33$, $\xi_4 = 64/111$. The OEIS entry A262957 [9], contributed by Mohamed Sabba in November 2015, lists the numerators $a(n)$ of ξ_n in lowest terms, with offset 1; the denominators form A263295 [10], and the analogous fractions beginning with $1+\cdots$ form A244279 and A244280 [7, 8]. The entry also records that ξ_n converges to $0.57663338973\ldots$ (A346590 [11]), a number with simple continued fraction $[0; 1, 1, 2, 1, 3, 4, 1, 5, 6, 1, 7, 8, \ldots]$. This follows from Peter Bala's note [2], linked from the entry, which shows that for every integer sequence (a_n) with $a_n > 1$ for $n \geq 2$ the infinite alternating continued fraction $a_1 - 1/(a_2 + 1/(a_3 - \cdots))$ converges, and gives its simple continued fraction explicitly in terms of the a_n.

On December 22, 2015, Robert Israel added to the entry a three-term recurrence with rational coefficients for sequences $b(n), c(n)$ such that $a(n)$ is the numerator of $b(n)/c(n)$, and conjectured that $b(n)$ and $c(n)$ are coprime for all n, "so that $a(n) = b(n)$"; he verified this for $n \leq 10000$ (the comment is quoted in Section 2). We prove the conjecture, including the integrality of $b(n)$ and $c(n)$ (Theorem 2); along the way we prove the identity $\xi_n = b(n)/c(n)$, which underlies Israel's assertion, stated in the entry without proof, that $a(n)$ is the numerator of $b(n)/c(n)$.

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Method. Let P_k/Q_k be the convergents of $f_1(f_2(f_3(\cdots)))$, so that $P_k = (k+1)P_{k-1} + (-1)^k P_{k-2}$, and likewise for Q_k. Since ξ_n is this continued fraction cut after the partial denominator $n+1$ with tail $(-1)^{n+1}/1$, we have $\xi_n = x(n)/y(n)$ with $x(n) = P_n + (-1)^{n+1}P_{n-1}$ and $y(n) = Q_n + (-1)^{n+1}Q_{n-1}$ (Lemma 4). A computation split by the parity of n shows that x and y satisfy Israel's recurrence (Lemma 6); the initial values are $2, 3$ and $3, 5$, so $b = x$ and $c = y$. Finally $x(n)Q_{n-1} - P_{n-1}y(n) = P_nQ_{n-1} - P_{n-1}Q_n = \pm 1$, so $x(n)$ and $y(n)$ are coprime, and they are positive.

Status and prior work. The statement is still marked "Conjecture" in A262957 (revision #80, March 4, 2025). Bala's note [2] concerns only the limit of ξ_n; it does not mention Israel's recurrence or the coprimality of $b(n)$ and $c(n)$, and the cross-referenced entries [10, 11, 7, 8] contain no proof. A262957 is not among the conjectures of the OEIS Open benchmark [1], nor in the GitHub repositories `epoch-research/LeanOpenProblems` and `LeanOpenProblems-results`, nor in formal-conjectures [3]; searches of arXiv, MathOverflow, Mathematics Stack Exchange and the SeqFan mailing list by the math0 referees found no proof or counterexample. We therefore classify the result as *new*, with the caveat that the argument is a routine application of the classical theory of convergents (see e.g. [4]); the one step that needs an idea is to recognize Israel's recurrence as the recurrence satisfied by $P_n \pm P_{n-1}$. The conjecture does not follow from the recurrence directly: the recurrence divides by $n \mp 1$, so integrality is not evident, and the cross-determinants $b(n+1)c(n) - b(n)c(n+1)$ have absolute value n or $n+2$, which, granting integrality, only shows that $\gcd(b(n), c(n))$ divides 3 (Remark 12).

2. Statement

Definition 1. For $n \geq 0$ let $\xi_n \in \mathbb{Q}$ be as above, and for $n \geq 1$ let $a(n)$ be the numerator of ξ_n in lowest terms. (For $k \geq 2$ and $t \geq 1$ we have $f_k(t) \geq k - 1 \geq 1$, so every inner value in ξ_n is at least 1 and no division by zero occurs; for $n \geq 1$, $\xi_n = 1 - 1/t$ with $t = f_2(\cdots) > 2$, so $\xi_n > 0$.) Define $b, c\colon \{1, 2, \dots\} \to \mathbb{Q}$ by $b(1) = 2$, $b(2) = 3$, $c(1) = 3$, $c(2) = 5$ and, for $n \geq 2$ and $w \in \{b, c\}$,

$$w(n+1) = \frac{\big((-1)^n(n-1) + n(n+2)\big)\, w(n) - \big(1 + (-1)^n(n+1)\big)\, w(n-1)}{n - (-1)^n}. \tag{1}$$

The divisor $n - (-1)^n \in \{n-1, n+1\}$ is nonzero for $n \geq 2$.

The COMMENTS section of A262957 [9] contains, verbatim:

```
From _Robert Israel_, Dec 22 2015: (Start)
a(n) is the numerator of b(n)/c(n) where
b(1) = 2, b(2) = 3, c(1) = 3, c(2) = 5,
b(n+1) = (((-1)^n*(n-1)+n*(n+2))*b(n) - (1+(-1)^n*(n+1))*b(n-1))/(n-(-1)^n),
c(n+1) = (((-1)^n*(n-1)+n*(n+2))*c(n) - (1+(-1)^n*(n+1))*c(n-1))/(n-(-1)^n).
Conjecture: b(n) and c(n) are coprime for all n, so that a(n) = b(n).
I have verified this for n <= 10000. (End)
```

Theorem 2. *For every $n \geq 1$, the numbers $b(n)$ and $c(n)$ are integers, $\gcd(b(n), c(n)) = 1$, and $a(n) = b(n)$.*

Theorem 2 is exactly the Lean statement below. The proof shows moreover that $b(n) > 0$, $c(n) > 0$ and $\xi_n = b(n)/c(n)$ for $n \geq 1$ (Corollaries 5 and 7 and Lemma 11); these facts are proved in the Lean file as auxiliary lemmas but are not part of the formal statement.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A262957.lean`, reviewed for fidelity and frozen (SHA-256 `a21c48e2...0895207`) before any proof attempt. Its declarations are, verbatim with the docstrings omitted:

```
import Mathlib
namespace Math0.Statements.A262957
def cf (n : ℕ) : ℚ :=
  (List.range' 1 (n + 1)).foldr (fun (k : ℕ) (t : ℚ) => (k : ℚ) + (-1) ^ k / t) 1
def a (n : ℕ) : ℕ := (cf n).num.natAbs
theorem a_data : (List.range 22).map (fun i => a (i + 1)) =
    [2, 3, 19, 64, 538, 2833, 29169, 210308, 2572158, 23595915, 334778571, 3732092084,
      60305234822, 791741083537, 14359827157009, 217037153818264, 4366918714540522,
      74685204276602819, 1651116684587556019, 31524723785455714840, 759659139498065625218,
      16017463672140861567617] := by
  decide +kernel
def step (m : ℕ) (u v : ℚ) : ℚ :=
  (((-1) ^ m * ((m : ℚ) - 1) + (m : ℚ) * ((m : ℚ) + 2)) * v
      - (1 + (-1) ^ m * ((m : ℚ) + 1)) * u) / ((m : ℚ) - (-1) ^ m)
def israelPair (x₁ x₂ : ℚ) : ℕ → ℚ × ℚ
  | 0 => (x₁, x₂)
  | n + 1 =>
    let p := israelPair x₁ x₂ n
    (p.2, step (n + 2) p.1 p.2)
def b (n : ℕ) : ℚ := (israelPair 2 3 (n - 1)).1
def c (n : ℕ) : ℚ := (israelPair 3 5 (n - 1)).1
theorem bc_data : (List.range 6).map (fun i => (b (i + 1), c (i + 1))) =
    [(2, 3), (3, 5), (19, 33), (64, 111), (538, 933), (2833, 4913)] := by
  decide +kernel
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n →
    ∃ p q : ℤ, b n = (p : ℚ) ∧ c n = (q : ℚ) ∧ Int.gcd p q = 1 ∧
      ((a n : ℕ) : ℚ) = b n
end Math0.Statements.A262957
```

Encoding. `List.range' 1 (n + 1)` is $[1, 2, \dots, n+1]$, so `cf n` is the right fold $f_1(f_2(\cdots f_{n+1}(1)\cdots)) = \xi_n$, computed exactly in $\mathbb{Q}$; as every inner value is at least 1, Lean's convention $x/0 = 0$ never intervenes. `Rat.num` is the numerator in lowest terms (with positive denominator), so `a n` $= |\mathrm{num}(\xi_n)| = a(n)$; the absolute value mirrors the entry's PARI program, which returns `denominator(1/t)`. `step m u v` is the right-hand side of (1) with $n = m$, $w(n-1) = u$, $w(n) = v$, and by induction `israelPair` $x_1\ x_2$ `k` is $(w(k+1), w(k+2))$ for the sequence with $w(1) = x_1$, $w(2) = x_2$ and (1), the step from k to $k+1$ using (1) at $n = k+2 \geq 2$. Hence `b n`, `c n` are $b(n)$, $c(n)$ for $n \geq 1$ (the values at $n = 0$ are unused). The data lemmas are proved by evaluation in the kernel (`decide +kernel`, which, unlike `native_decide`, does not trust compiled code): `a_data` reproduces all 22 terms $a(1), \dots, a(22)$ listed in A262957, and `bc_data` gives $(b(n), c(n))$ for $n \leq 6$, matching the initial values, the examples 19/33 and 64/111, and the first terms of A263295.

Reading of the conjecture. Since (1) divides, $b(n)$ and $c(n)$ are a priori rationals; "coprime" presupposes that they are integers, and `statement` asserts this (the integers p, q) together with $\gcd(p, q) = 1$. It keeps the stated consequence "so that $a(n) = b(n)$" as a conjunct, cast to $\mathbb{Q}$; this depends on Israel's unproved assertion that $a(n)$ is the numerator of $b(n)/c(n)$, which our proof establishes (Corollaries 5 and 7). The range $n \geq 1$ is the index range of b and c.

3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A262957.lean`; each step names the corresponding Lean declaration.

Definition 3 (`seq`, `z`)**.** For $r_0, r_1 \in \mathbb{Z}$ let $r\colon \mathbb{N} \to \mathbb{Z}$ be given by $r(0) = r_0$, $r(1) = r_1$ and

$$r(k+2) = (k+2)\,r(k+1) + (-1)^{k+1}\,r(k) \qquad (k \ge 0), \tag{2}$$

and put $z_r(n) = r(n+1) + (-1)^{n+1} r(n)$. Let p and q be the sequences with $(r_0, r_1) = (1,1)$ and $(0,1)$, and put $x = z_p$, $y = z_q$.

Thus $p = 1,1,1,4,15,79,459,\dots$, $q = 0,1,2,7,26,137,796,\dots$, $x(1), x(2), \dots = 2,3,19,64,\dots$ and $y(1), y(2), \dots = 3,5,33,111,\dots$. In the notation of the introduction $p(k) = P_{k-1}$ and $q(k) = Q_{k-1}$, with the usual initial values $P_{-1} = P_0 = 1$, $Q_{-1} = 0$, $Q_0 = 1$ [4]; we do not use this interpretation, since Lemma 4 proves what is needed.

Lemma 4 (`fold_eq`)**.** *For every $k \ge 0$ and every $t \in \mathbb{Q}$ with $t \ge 1$,*

$$f_1\big(f_2(\cdots f_{k+1}(t)\cdots)\big) = \frac{p(k+1)\,t + (-1)^{k+1} p(k)}{q(k+1)\,t + (-1)^{k+1} q(k)}. \tag{3}$$

As in Lean, $u/0$ is read as 0; the only property of division used is $(N/t)/(D/t) = N/D$ for $t \neq 0$, which also holds when $D = 0$. The left-hand side involves no division by zero, since every argument of an f_i is at least 1.

Proof. Induction on k, for all $t \ge 1$ simultaneously. Write $F_k(t)$ for the left-hand side and $\varepsilon_k = (-1)^k$. For $k = 0$, $F_0(t) = 1 - 1/t = (t-1)/t$, which is the right-hand side $(1\cdot t - 1)/(1\cdot t - 0)$. Let $k \ge 0$ and $t \ge 1$. Since $[1,\dots,k+2]$ is $[1,\dots,k+1]$ followed by $k+2$ (in Lean, `List.range'_concat` and `List.foldr_append`), $F_{k+1}(t) = F_k(t')$ with $t' = f_{k+2}(t) = k + 2 + \varepsilon_{k+2}/t$. As $0 < 1/t \le 1$, $t' \ge k+1 \ge 1$, so the induction hypothesis gives $F_{k+1}(t) = N'/D'$ with $N' = p(k+1)t' + \varepsilon_{k+1} p(k)$ and $D' = q(k+1)t' + \varepsilon_{k+1} q(k)$. By (2),

$$t\,N' = \big((k+2)p(k+1) + \varepsilon_{k+1} p(k)\big)t + \varepsilon_{k+2}\,p(k+1) = p(k+2)\,t + \varepsilon_{k+2}\,p(k+1) =: N,$$

and likewise $t\,D' = q(k+2)t + \varepsilon_{k+2} q(k+1) =: D$. Hence $F_{k+1}(t) = (N/t)/(D/t) = N/D$, which is (3) for $k+1$. □

Corollary 5 (`cf_eq`)**.** *For every $n \ge 0$, $\xi_n = x(n)/y(n)$.*

Proof. Take $k = n$ and $t = 1$ in Lemma 4. □

Lemma 6 (`step_z`)**.** *Let r satisfy (2) and $z = z_r$. For every $m \ge 0$, z satisfies (1) at $n = m+2$:*

$$z(m+3) = \frac{\big((-1)^m(m+1) + (m+2)(m+4)\big)\,z(m+2) - \big(1 + (-1)^m(m+3)\big)\,z(m+1)}{m + 2 - (-1)^m}.$$

Proof. Put $u = r(m+1)$, $v = r(m+2)$ and $s = (-1)^m$, so $s^2 = 1$. By (2), $r(m+3) = (m+3)v + su$ and $r(m+4) = (m+4)\,r(m+3) - sv$, hence

$$z(m+1) = v + su, \qquad z(m+2) = (m+3-s)\,v + su,$$
$$z(m+3) = \big((m+4+s)(m+3) - s\big)v + \big(1 + (m+4)s\big)u.$$

The divisor $m + 2 - s \in \{m+1, m+3\}$ is nonzero, so the claim is equivalent to

$$\big(s(m+1) + (m+2)(m+4)\big)\,z(m+2) - \big(1 + s(m+3)\big)\,z(m+1) = (m+2-s)\,z(m+3).$$

If $s = 1$, both sides equal $(m+1)(m+5)\,u + (m^3+9m^2+22m+14)\,v$: the left is $(m^2+7m+9)\big(u + (m+2)v\big) - (m+4)(u+v)$ and the right is $(m+1)\big((m+5)u + (m^2+8m+14)v\big)$. If $s = -1$, both sides

equal $-(m+3)^2u+(m^3+9m^2+28m+30)\,v$: the left is $(m^2+5m+7)\big((m+4)v-u\big)+(m+2)(v-u)$ and the right is $(m+3)\big(-(m+3)u+(m^2+6m+10)v\big)$. □

In Lean, after rewriting $r(m+3)$ and $r(m+4)$ by (2), the proof splits on $(-1)^m = \pm 1$ (`neg_one_pow_eq_or`), clears the divisor (`div_eq_iff`) and closes each case with `ring`.

Corollary 7 (`israelPair_z`)**.** *For every $n \geq 1$, $b(n) = x(n)$ and $c(n) = y(n)$; in particular $b(n), c(n) \in \mathbb{Z}$.*

Proof. We have $x(1) = p(2) + p(1) = 2$, $x(2) = p(3) - p(2) = 3$, $y(1) = q(2) + q(1) = 3$ and $y(2) = q(3) - q(2) = 5$, the initial values of b and c. For r satisfying (2), induction on k with Lemma 6 at $m = k$ shows that `israelPair` started from $(z_r(1), z_r(2))$ equals $(z_r(k+1), z_r(k+2))$ at step k. Taking $r = p, q$ and $k = n - 1$ gives the claim. □

Lemma 8 (`det_eq`)**.** *For every $k \geq 0$, $d(k) := p(k+1)q(k) - p(k)q(k+1) \in \{1, -1\}$.*

Proof. $d(0) = 1 \cdot 0 - 1 \cdot 1 = -1$, and by (2), with $\varepsilon = (-1)^{k+1}$,

$$d(k+1) = \big((k+2)p(k+1) + \varepsilon p(k)\big)q(k+1) - p(k+1)\big((k+2)q(k+1) + \varepsilon q(k)\big) = -\varepsilon\, d(k).$$ □

Corollary 9. *For every $n \geq 0$, $\gcd(x(n), y(n)) = 1$.*

Proof. The terms $\pm p(n)q(n)$ cancel in $x(n)q(n) - p(n)y(n)$, which therefore equals $p(n+1)q(n) - p(n)q(n+1) = d(n)$. With $d = d(n)$ we have $d^2 = 1$ by Lemma 8, so $\big(d\,q(n)\big)x(n) + \big(-d\,p(n)\big)y(n) = 1$. In Lean these coefficients witness `IsCoprime (z 1 1 n) (z 0 1 n)` (closed by `linear_combination`), which `Int.isCoprime_iff_gcd_eq_one` turns into gcd = 1. □

Lemma 10 (`seq_mono`)**.** *If $0 \leq r_0 \leq r_1$, then $0 \leq r(k) \leq r(k+1)$ for every $k \geq 0$.*

Proof. Induction on k. If $0 \leq r(k) \leq r(k+1)$, then $r(k+1) \geq 0$ and $(-1)^{k+1}r(k) \geq -r(k) \geq -r(k+1)$, so by (2) $r(k+2) \geq (k+1)r(k+1) \geq r(k+1)$. □

Lemma 11 (`z_pos`)**.** *If $0 \leq r_0 \leq r_1$ and $r_1 \geq 1$, then $z_r(n) > 0$ for every $n \geq 1$. In particular $x(n), y(n) > 0$ for $n \geq 1$.*

Proof. Write $n = j + 1$ and $s = (-1)^j$. By Lemma 10, r is nondecreasing, so $r(j+1) \geq r_1 \geq 1$ and $0 \leq r(j) \leq r(j+1)$. By (2), $z_r(j+1) = r(j+2) + s\,r(j+1) = (j+2+s)\,r(j+1) - s\,r(j)$. If $s = 1$ this is $(j+3)r(j+1) - r(j) \geq (j+2)r(j+1) > 0$; if $s = -1$ it is $(j+1)r(j+1) + r(j) > 0$. The pairs $(r_0, r_1) = (1, 1)$ and $(0, 1)$ satisfy the hypotheses. □

Proof of Theorem 2 (`result`). Let $n \geq 1$. By Corollary 7, $b(n) = x(n)$ and $c(n) = y(n)$ are integers; they are positive by Lemma 11, coprime by Corollary 9, and $\xi_n = x(n)/y(n)$ by Corollary 5. A fraction x/y with $y > 0$ and $\gcd(x, y) = 1$ has numerator x in lowest terms (`Rat.num_div_eq_of_coprime`), so $a(n) = |x(n)| = x(n) = b(n)$. In the Lean statement take $p = x(n)$, $q = y(n)$. □

Remark 12 (not formalized)**.** Applied to (1) directly, the determinant argument does not suffice. Let $D(n) = b(n+1)c(n) - b(n)c(n+1)$. Substituting (1), the terms in $b(n)c(n)$ cancel and $D(n) = \frac{1+(-1)^n(n+1)}{n-(-1)^n} D(n-1)$ for $n \geq 2$, with $D(1) = -1$; the factor is $(n+2)/(n-1)$ for even n and $-n/(n+1)$ for odd n, so $|D(n)| = n$ for odd n and $n+2$ for even n ($D = -1, -4, 3, 6, -5, -8, \dots$). Granting integrality, a common divisor of $b(n)$ and $c(n)$ divides $D(n-1)$ and $D(n)$; for odd $n \geq 3$ these are $\pm(n+1)$ and $\pm n$, but for even n they are $\pm(n-1)$ and $\pm(n+2)$, so a common factor 3 is not excluded when $n \equiv 4 \pmod 6$ (e.g. $n = 4$: $D(3) = 3$, $D(4) = 6$, and $3 \mid c(4) = 111$).

4. Machine verification

The proof is the file `lean/Math0/Frontier/OEIS/A262957.lean` of the math0 repository (`https://github.com/Foundation-0/math0`). It imports the frozen statement file of Section 2 and the audit commands `lean/Math0/Audit.lean`. The main declaration is

```
theorem result : statement
```

in namespace `Math0.Frontier.A262957`, with type `Math0.Statements.A262957.statement`. All auxiliary declarations are `private`: `seq`, `seq_succ_succ`, `z`, `step_z`, `israelPair_z`, `seq_mono`, `z_pos`, `det_eq`, `fold_eq` and `cf_eq`, as in Section 3; p, q, x, y are `seq 1 1`, `seq 0 1`, `z 1 1`, `z 0 1`. The proof treats all n at once; the only evaluations are of $z_p(1), z_p(2), z_q(1), z_q(2)$ (by `norm_num`) and of $d(0) = -1$ (by `decide`). The data lemmas `a_data` and `bc_data` are not used. The file contains no `sorry`, `native_decide`, `set_option` or `axiom`.

Audit. The file ends with

```
#math0_settles Math0.Frontier.A262957.result Math0.Statements.A262957.statement
```

This command, used for statements that math0 formalizes itself, fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, the `Lean.ofReduceBool` and `Lean.trustCompiler` axioms behind `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the constant `statement` or its negation, up to metadata annotations, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A262957.lean` in `lean/` prints

```
math0 audit: 'Math0.Statements.A262957.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A262957.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A262957.result' proves 'Math0.Statements.A262957.statement'
```

Toolchain and reproduction. Lean 4 [6] `v4.33.1` and Mathlib [5] `v4.33.1`, pinned in `lean/lean-toolchain` and `lean/lake-manifest.json`; the repository also pins formal-conjectures [3] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root. It fetches the Mathlib build cache, builds all math0 results with `lake build` (which re-runs every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks.

References

[1] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941, 2026.

[2] P. Bala, *A note on A262957 and A263295*, December 24, 2015, linked from [9]: `https://oeis.org/A262957/a262957.pdf`.

[3] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. `https://github.com/google-deepmind/formal-conjectures`.

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